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An Optimization Method for Manned/Unmanned Aerial Vehicle Collaborative Operation System Architecture Based on PGQNSGA-II

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Abstract: The architecture of the Manned/Unmanned Aerial Vehicle Collaborative Operation System (MAV/UAV COS) is crucial for improving combat effectiveness and resource utilization efficiency. Optimizing this architecture involves managing complex, interdependent components, which presents a constrained multi-objective optimization challenge. Initially, the elements of the MAV/UAV COS architecture were analyzed and formally expressed, transforming the architecture optimization problem into a multi-objective optimization problem, with the objectives of maximizing total system effectiveness, command-and-control performance, and system execution performance. Constraints were formulated based on mission and payload information. Subsequently, a Quantum Non-Dominated Sorting Genetic Algorithm based on Preference Guidance (PGQNSGA-II) was developed, incorporating an adaptive quantum gate mechanism based on preference information to enhance chromosome updating, ensuring that the probability amplitude of quantum bits aligns more closely with the optimal chromosome. The simulation results demonstrate that the proposed PGQNSGA-II algorithm significantly enhances the global search capability and efficiency compared to traditional quantum genetic algorithms, making it well-suited for optimizing MAV/UAV COS architectures.

Keywords: unmanned aerial vehicle; manned aerial vehicle; architecture; optimization; quantum



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1. Introduction

Architecture is the core of the Manned/Unmanned Aerial Vehicle Collaborative Operation System (MAV/UAV COS), playing a crucial role throughout its entire lifecycle, including requirements analysis, system design, development, testing, and deployment [1]. Optimizing the system architecture can help to enhance combat effectiveness and resource utilization efficiency. This process involves the integrated optimization of multiple aspects, including combat missions, resource allocation, system functionality, communication networks, and command and control, thereby enhancing the overall combat capability and adaptability of the entire system [2]. However, optimizing the architecture of MAV/UAV COS is complex due to the numerous and interdependent components. These interactions affect multiple objectives simultaneously, framing optimization as a constrained multi-objective optimization problem. Unlike single-objective optimization problems [3,4], multi-objective optimization problems do not have a single optimal solution, but instead yield a set of feasible solutions that balance competing objectives [5]. Due to their ease of implementation and global search capabilities, multi-objective heuristic evolutionary algorithms are widely employed for solving such optimization problems. Currently, there is limited research on the multi-objective optimization of MAV/UAV COS architectures. Shu et al. [6] developed an optimization model for MAV/UAV COS from the perspectives of mission, equipment, and command and control (C2) structure, using the P-NSGA-III algorithm to solve it. Zhou et al. [7] employed the MOPSO optimization algorithm combined with machine learning techniques to identify key performance indicators for evaluating

MAV/UAV COS effectiveness. Han et al. [8] introduced and validated a multi-objective hybrid swarm algorithm founded on Holon organizational theory, designed to address the multi-objective optimization of coalition formation among manned/unmanned aerial vehicle swarms in collaborative combat scenarios.

Current research on these algorithms focuses on improving Pareto frontiers and reducing search times by enhancing solution convergence and propagation methods. In this context, the integration of quantum theory introduces a novel approach to addressing these challenges. Quantum algorithms are characterized by their significant computational speed, particularly in handling optimization problems with numerous computational elements, aligning well with the speed requirements of online platforms for MAV/UAV COS architecture. Since 1992, when the first quantum algorithm was proposed [9], researchers have gradually developed some important quantum algorithms. Among these, the quantum genetic algorithm is a combination of quantum computing theory and the traditional genetic algorithm, which has the advantages of the traditional genetic algorithm and the excellent characteristics of quantum computing: computational speed block, low computational complexity, and outstanding optimization ability. In recent years, experts have begun to explore the existing quantum genetic algorithms for solving multi-objective optimization problems. Olvera et al. [10] improved two quantum heuristic algorithms and proposed a new algorithm. They proved that these algorithms perform better on problems with constraints and complex Pareto sets compared with the Non-Dominated Sorting Genetic Algorithm (NSGA-II), providing valuable results for future research on multi-objective optimization in quantum computers. Valuable results were provided by He et al. [11], who proposed a multi-objective quantum genetic algorithm for solving building envelope optimization models and showed strong search and optimization capabilities compared to NSGA-II. Abraham et al. [12] proposed a quantum-inspired evolutionary algorithm that combines NSGA-II fast non-dominated sorting and congestion distance through a distributed implementation, which enables the algorithm to show excellent performance in terms of hypervolume and Δ' metrics.

Although quantum algorithms offer significant computational speed advantages, most studies focus on single-objective optimization problems due to the decoherence of qubit operators in multi-objective optimization, which often leads to premature convergence and local optima [5]. In response, this chapter proposes a Quantum Non-Dominated Sorting Genetic Algorithm based on Preference Guidance (PGQNSGA-II), which adaptively adjusts quantum gate rotation angles based on the differences between target vectors and preference information. The preference information, tailored to the characteristics of manned/unmanned aerial vehicle formations, guides the population toward the optimal direction, thereby mitigating the premature convergence issue in traditional multi-objective quantum genetic algorithms. First, this study analyzes and selects appropriate optimization objectives and decision variables to establish a mathematical model of the optimization problem. Based on the characteristics of the decision variables, a quantum encoding strategy is proposed, along with a preference-guided quantum genetic algorithm to solve the optimization objective function for the MAV/UAV COS architecture. Finally, simulation experiments are conducted to evaluate the proposed optimization algorithm's effectiveness and advantages in terms of computational speed, accuracy, and performance in specific MAV/UAV COS interception combat scenarios.

2. Optimization Model

2.1. Mathematical Description and Related Definitions

System architecture optimization enhances overall combat capability and efficiency by strategically designing and adjusting the components of the combat system. Optimizing combat system architecture involves multiple levels, including tactics, organizational structure, resource allocation, and other critical aspects. The mathematical formulation and relevant definitions of the system architecture optimization problem are provided below.

Definition 1. *Architecture optimization problem.*

For Definition 1, $f(z)$ denotes a multi-objective system architecture optimization function characterized by non-linearity and multi-modality, where y is the number of objectives to be optimized and n is the number of decision variables z . The mathematical description of the architecture optimization problem is given as follows [13]:

$$\min F(z) = (f_1(z), f_2(z), f_3(z), \dots, f_y(z))^T \quad (1)$$

$$\text{s.t. } -\infty < z_i^{lb} < z_i < z_i^{up} < \infty \forall i = 1, 2, \dots, n \quad (2)$$

$$z = (z_1, z_2, \dots, z_n)^T, z \in Z, Z \in \mathbb{R}^n \quad (3)$$

where $F(z)$ represents the set of objective functions, z is the n -dimensional decision variable, Z is the set of decision variables, and $\mathbb{R}^n$ is the decision space. z_i^{lb} and z_i^{up} represent the upper and lower bounds of the decision variables z_i , $\forall i = 1, 2, \dots, n$. Due to the default conflicting nature among the objectives in a multi-objective optimization problem, the solution of a multi-objective optimization problem is defined by the Pareto optimal solution set.

Definition 2. *Pareto prevails [14].*

For two feasible solution vectors, $z_A = (z_{A1}, z_{A2}, \dots, z_{An})$ and $z_B = (z_{B1}, z_{B2}, \dots, z_{Bn})$, the solution vector z_A Pareto dominates the solution vector z_B , or z_A Pareto dominates z_B , so $z_A < z_B$, and when $\forall i \in \{1, 2, \dots, y\}$, there is $f_i(z_A) \leq f_i(z_B)$ and $\exists j \in \{1, 2, \dots, y\}$, and $f_j(z_A) < f_j(z_B)$ holds.

Definition 3. *Pareto optimal solution set [15].*

The set consisting of Pareto optimal solutions is called the Pareto optimal solution set. The set of multi-objective function images corresponding to the optimal solution set in the objective space constitutes the Pareto front surface.

2.2. Model Assumptions

Since the MAV/UAV COS will differ in various aspects, such as operation process, coordination mode, and load function, under the demands of different combat scenarios, the following assumptions are made:

Assumption 1. *The values for MAV operator skills and on-board load performance are fixed values and do not consider variations due to external factors.*

Assumption 2. *The MAV operator's command input (e.g., keying, touchscreen handwriting, or voice, etc.) time [16] is ignored in the calculation of his/her time to process the UAV task.*

Assumption 3. *The level of performance of the architecture to perform a task is a linear sum of the performance levels of the different loads, and synergies between the functional systems contained in the system can enhance the overall level of capability of the system.*

Assumption 4. *The interactions between drones or between a MAV and a drone resulting from synergistic behavior are set as synergistic relationships.*

Assumption 5. *In the calculation of system loads, operator loads and communication loads, which are affected by architectural elements, are mainly considered, and physical resource consumption is ignored.*

Assumption 6. *UAV payload performance follows the principle of Increasing Precision with Decreasing Intelligence (IPDI) [17].*

Assumption 7. *UAVs carry no more than one detection device of the same type.*

Assumption 8. In order to ensure the safety of MAVs, MAVs are not involved in attack missions, and all MAVs are isomorphic and carry remote detection equipment.

Based on the above assumptions of the research problem, the next step is to model the decision variables, objective functions and constraints in the architecture optimization problem, and to construct the architecture optimization model of MAV/UAV COS.

2.3. Architecture Element Calculus

Traditional combat system architecture elements typically encompass combat resources (e.g., personnel, equipment), command-and-control mechanisms, and tactical behaviors. Analyzing and standardizing the expression of combat system architecture elements are prerequisites for developing a specific system architecture optimization model. Therefore, this section provides definitions of MAV/UAV COS architecture elements and their formalized descriptions, as follows:

(1) Architecture resources

Architecture resources refer to the available equipment and personnel within the system, whose attributes include the quantity of resources, their functional capabilities, performance metrics, and the capacity of task loads executed.

The set of MAVs is $m_j \in M = \{m_1, m_2, \dots, m_{N_m}\}$ and N_m is the number of MAVs. Based on Assumption 8, the number of weapons carried by a MAV is $w_m = 0$.

The set of UAVs is $u_i \in U = \{u_1, u_2, \dots, u_{N_u}\}$, N_u is the number of UAVs, and the UAV u_i Detection Equipment Load Configuration Matrix $S_u = \{s_{u_i k_1}\}_{k_1 \times N_u}$ is denoted as

$$s_{u_i k_1} = \begin{cases} 1 & \text{if UAV } u_i \text{ carries } k_1 - \text{class sensors, } k_1 = 1, 2, \dots, N_s \\ 0 & \text{else} \end{cases} \quad (4)$$

UAVs u_i carry a total of N_w types of weapons. UAV Weapon Load Configuration Matrix:

$$W_u = \{w_{u_i k_2}\}_{k_2 \times N_u} \quad (5)$$

where $w_{u_i k_2}$ means that the drone u_i carries k_2 class of weapons $w_{u_i k_2}$ and $k_2 = 1, 2, \dots, N_w$. The intelligence level of the drone u_i is $LOA = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. The location coordinate of the UAV is $L_i = \{x_i, y_i, z_i\}$.

(2) Operational mission activity set

When the top-level mission is determined, the mission can be cascaded down to the smallest mission activity unit based on the situational information. There are N_t combat mission activity units, Act is recorded as the set of combat mission activities, the combat mission activity unit is recorded as Act_p , and then there is $Act_p \in Act = \{Act_1, Act_2, \dots, Act_{N_t}\}$. Record Act_p the location of the combat mission activity unit as $L_{Act_p} = \{x_{Act_p}, y_{Act_p}, z_{Act_p}\}$, the demand vector as $R_{Act_p} = \{s_1, s_2, \dots, s_{k_1}, w_1, w_2, \dots, w_{k_2}\}$, and d_{jip} indicates that the activity is executed by the UAV at u_i , and $D = \{d_{Act_p i}\}_{N_t \times N_u}$.

(3) Architectural relationships

With advancements in UAV intelligence, these systems exhibit high levels of autonomy and flexibility, and the degree of UAV intelligence directly influences the organizational structure of MAV and UAV interactions. The operation system architecture between MAV and UAV shown in Figure 1 can be viewed as a series of alignment strategies aimed at optimizing mission execution with limited resources. Based on Assumption 4, specific command-and-control relationships are classified into the following categories: ① direct control, ② delegation, ③ decision-making authorization, ④ supervision (primarily for unexpected events), and ⑤ full collaboration. These alignment strategies correspond directly to the UAV level of intelligence (LOI) [18], as detailed in Table 1.

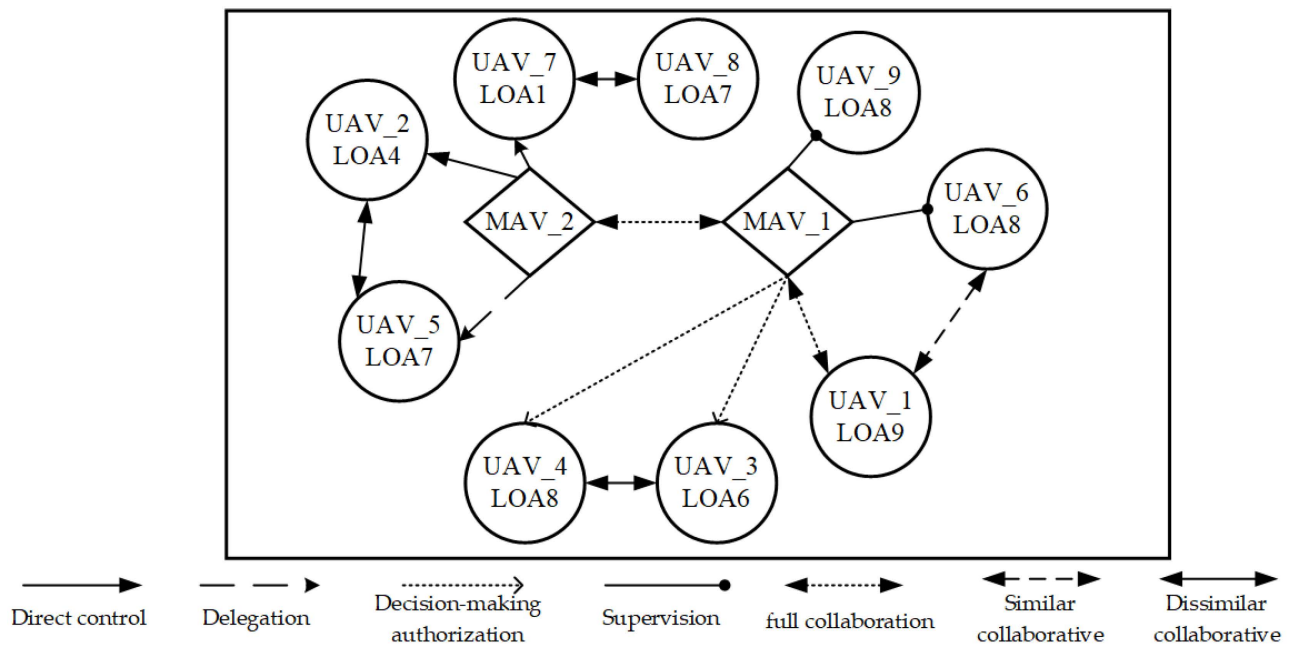


Figure 1. MAV/UAV COS architectures.

Table 1. Mapping of interoperability levels to command-and-control relationships and their automated descriptions.

LOI	Command-and-Control Relationship Levels	Automation Description
1	Direct control	UAV cannot autonomously execute operator commands
2	Delegation	The operator must send mission commands to the UAV, which can execute them autonomously
3	Decision-making authorization	The decision-making system proposes a reference proposal and, if approved, the UAV can autonomously execute the proposal
4	Supervision	The computer proposes an alternative and allows a human to veto it for a limited period of time before executing it automatically (exception management), and the UAV can execute the proposal autonomously.
5	Full collaboration	Operators are not involved in decision-making, intelligent systems make decisions and execute autonomously

The inter-UAV synergistic relationships include the following: ① similar collaborative relationships, in which similar actuators act on the same object, ② dissimilar collaborative relationships, in which dissimilar actuators act on the same object, and ③ no collaborative relationships, where actuators act on different objects. The MAV–UAV inter-organizational structure $RE_1 = \{r_{ijl_1}\}_{N_u \times N_m}$ and the UAV–UAV inter-organizational structure $RE_2 = \{r_{ijl_2}\}_{N_u \times N_u}$ are represented as follows:

$$re_{ijl_1} = \begin{cases} 1 & \text{if the collaborative relationship between UAV}_i \text{ and MAV}_j \text{ is } l_1, \\ 0 & \text{else} \end{cases} \quad (6)$$

$$re_{ijl_2} = \begin{cases} 1 & \text{if the collaborative relationship between UAV}_i \text{ and UAV}_j \text{ is } l_2, \\ 0 & \text{else} \end{cases} \quad (7)$$

2.4. Objective Function

The multi-objective function in the optimization problem of MAV/UAV COS architecture can be decomposed into the following three objective functions:

(1) Overall system effectiveness

The overall effectiveness of the MAV/UAV COS is obtained from the calculations in the Human Factors Availability–Dependability–Capability (HFADC) methodology [19],

which is a comprehensive effectiveness assessment methodology that takes into account manned-aircraft-operator-related performance and is as follows for a team of N_m MAVs and N_u UAVs:

$$f_e = \prod_{j=1}^{N_m} ADC_j \quad (8)$$

(2) System command-and-control performance

The command-and-control capability of MAV/UAV COS includes the cost of MAV decision load $f_{Lo_{mu}}$ and the level of MAV command and control $f_{Per_{mu}}$, i.e., the decision load of MAV command and control of UAVs. Based on Assumption 5, $f_{Lo_{mu}}$ is related to the number of UAVs and the synergistic relationship, and $load_{l_1}$ denotes the decision load of MAV command and control of UAVs whose synergistic relationship is l_1 , denoted as

$$f_{Lo_{mu}} = \sum_{j=1}^{N_m} \sum_{i=1}^{N_u} re_{ijl_1} \cdot load_{l_1} \quad (9)$$

MAV command and control of UAV performance is related to the inherent skill level of the MAV operator, the number of UAVs, and the synergistic relationship. per_{l_1} denotes the performance value of MAV command and control of UAVs with a synergistic relationship of l_1 [20], denoted as

$$f_{Per_{mu}} = \sum_{j=1}^{N_m} \sum_{i=1}^{N_u} re_{ijl_1} \cdot per_{l_1} \quad (10)$$

In summary, the system command-and-control performance is

$$f_m(RE_1) = f_{Per_{mu}} / f_{Lo_{mu}} \quad (11)$$

(3) System implementation performance

The calculation of mission execution capability of command and control of UAVs includes UAV mission execution performance f_{Per_u} and load cost of cooperative communication between UAVs $f_{Lo_{com}}$. Among these factors, the UAV mission execution performance is related to the UAV load performance level per_k and the intelligence level, based on Assumption 6. The load accuracy Acc decreases with the increase in the UAV intelligence level, and information entropy is used to establish the influence operator between the UAV intelligence level and the load accuracy Acc_{u_i} [6]:

$$Acc_{u_i} = e^{-\xi \cdot LOA_{u_i}} \quad (12)$$

where ξ is the coefficient of influence of intelligence level on accuracy and LOA_{u_i} is the UAV intelligence level. Therefore, f_{Per_u} is denoted as

$$f_{Per_u}(S_u, W_u, LOA_{ui}) = \sum_{k_1=1}^{N_s} \sum_{i=1}^{N_u} per_{k_1} \cdot (s_{u_i k_1} \cdot Acc_{u_i}) + \sum_{k_2=1}^{N_w} \sum_{i=1}^{N_u} per_{k_2} \cdot (w_{u_i k_2} \cdot Acc_{u_i}) \quad (13)$$

where, based on Assumption 1, per_{k_1} and per_{k_2} denote the intrinsic performance values of the loads, respectively. The communication load cost $f_{Lo_{com}}$ is related to the synergistic relationship between UAVs, and $load_{l_2}$ denotes the communication load for which the synergistic relationship between UAVs is l_2 , denoted as

$$f_{Lo_{com}}(RE_2) = \sum_{j=1}^{N_u} \sum_{i=1}^{N_u} re_{ijl_2} \cdot load_{l_2} \quad (14)$$

In summary, the system's ability to perform its tasks is

$$f_u(S_u, W_u, LOA_{ui}, RE_2) = f_{Per_u} / f_{Lo_{com}} \quad (15)$$

In addition, the optimization model of the MAV/UAV cooperative warfare architecture should also satisfy the following constraints to ensure that the mission can be accomplished efficiently and safely, as follows:

$$\forall i \in I, \sum_{k_1 \in N_s} s_{u_i k_1} + \sum_{k_2 \in N_w} w_{u_i k_2} \geq 1 \quad (16)$$

$$Act_p \in Act, R_{Act_p} \subset \cup_{i \in I} (S_{u_i} \cup W_{u_i}) \cdot d_{Act_p i} \quad (17)$$

$$\forall Act_p \in Act, R_{Act_p} \neq \emptyset \quad (18)$$

$$Act_p \in Act, \sum_{i \in I} d_{Act_p i} \geq 1 \quad (19)$$

$$\forall j \in J, \sum_{i \in I} re_{ijl_1} \geq 1 \quad (20)$$

Constraint (16) ensures that each UAV is equipped with at least one detection sensor or weapon. Constraint (17) ensures that the system meets the necessary functional requirements for each mission activity unit. Constraint (18) stipulates that each mission activity unit must have at least one functional payload to operate. Constraint (19) ensures that each mission is assigned at least one UAV for execution. Constraint (20) ensures that each MAV maintains command and control over at least one UAV.

In summary, the optimization problem for the MAV/UAV COS architecture aims to maximize overall system effectiveness, command-and-control performance, and execution performance, all within the bounds of multiple constraints.

3. PGQNSGA-II Optimization Algorithm

3.1. Algorithmic Framework

The Quantum Non-Dominated Sorting Genetic Algorithm (QNSGA-II) is an optimization approach that integrates the benefits of quantum computing with NSGA-II. NSGA-II is an optimization algorithm based on natural selection and genetic mechanisms, while the Quantum Genetic Algorithm enhances NSGA-II's global search capability and convergence through quantum computing concepts such as superposition states and quantum gate operations. To further enhance convergence speed to meet the optimization requirements of online platforms for MAV/UAV COS, this study proposes PGQNSGA-II by refining the quantum gate operation strategy.

Algorithm 1 provides the pseudo-code of the PGQNSGA-II algorithm, which initializes the population by randomly generating the initial quantum population, where each individual is represented by a quantum bit, and the quantum state is initialized as a uniform superposition state. The coordinates of the preference point and the maximum number of iterations are set, allowing the algorithm to evolve iteratively for the specified number of generations. The measurement yields the collapsed state $K(0)$, which is then used to evaluate its fitness. In each generation, a new quantum gate operation, guided by evolutionary operations and preference points, is performed on the selected parent individual to generate a new candidate solution, and the populations are merged. The combined quantum population is measured, mapping the quantum states to classical states, followed by calculating the fitness of each solution and determining its dominance relation and rank. The top n solutions are selected as the new parent populations. This iterative process is repeated until the stopping condition, such as reaching the maximum number of iterations, is met. The optimal solution to the architecture optimization problem is determined by calculating the Euclidean distance between the optimal solution set and the ideal point.

Algorithm 1. Pseudo-code of the PGQNSGA-II algorithm

Input: Initial candidate solutions: $P(0)$. Preferred point coordinates: V . Maximum number of iterations: $T_{\max gen}$. The scale of initial candidate solutions is N . The size of objectives is M . The adaptive rotation single parameters are $\theta_{\max}$ and $\theta_{\min}$.

Output: Global optimal vector b and its fitness fit

```

1   $it \leftarrow 0$ 
2   $P(it) = \text{initialize}(N)$ 
3  Obtain  $K(it)$  by measuring the states of  $P(it)$ 
4   $fit(K(it)) = \text{objective function}(K(it))$ 
5  Perform g-dominated sorting on  $K(it)$  and obtain all Pareto fronts  $P_f(it)$ 
6  while  $it < T_{\max gen}$  do
7     $it = it + 1$ 
8    for each individual  $i$  in the population  $P$  do
9      for each qubit  $j$  in individual  $i$  do
10       if qubit  $j$  differs then
11         Compute adaptive rotation angle  $\theta_i$ :  $\theta_i = \theta_{\max}((\frac{f(b_{ij}) - f(x_{ij})}{f(b_{ij})}) + (\frac{f(x_{ij}) - f(p_{ij})}{f(p_{ij})})) - (\theta_{\max} - \theta_{\min}) \frac{it}{T_{\max gen}}$ 
12         Adjust the quantum angle  $\theta(i, j)$  by:  $\theta(i, j) = \theta(i, j) + \theta_i * \text{rotation\_direction}(P_f[i][j])$ 
13         Ensure  $\theta(i, j)$  is within  $[0, 2\pi]$ :  $\theta(i, j) = \text{mod}(\theta(i, j), 2 \times \pi)$ 
14       end if
15     end for
16   end for
17   Update  $P(it - 1)$  with respect to  $P_f(it - 1)$  using self-adaption quantum rotation gate
18   Obtain  $K(it)$  by measuring the states of  $P(it - 1)$ 
19    $fit(K(it)) = \text{objective function}(K(it))$ 
20   Perform g-dominated sorting on  $K(it) \cup K(it - 1)$  and obtain all Pareto fronts  $F_t$ 
21   Form  $P(it)$  by accommodating the quantum individuals belonging to the different Pareto fronts starting from the best front
22 end while

```

3.2. Quantum Coding and Measurement

Before optimizing the architecture of the MAV/UAV COS, the population individuals must be encoded. In this study, the population individuals consist of the matrices RE_1 and RE_2 , which are converted from a matrix of variables to a vector of variables by extracting the matrix row vectors, as shown in Figure 2. For the MAV/UAV COS candidate, the vector dimension of the classical solution is denoted as $N = N_u \cdot N_m + N_u \cdot N_{ur}$, with the quantum bit length matching this dimension. Additionally, the number of quantum chromosomes is $2N$.

In quantum genetic algorithms, population individuals are represented by quantum chromosomes, and the chromosome genes are stored by each of the smallest units of information in quantum computing—quantum bits. A quantum bit can be in state 1 (denoted as $|1\rangle$), state 0 (denoted as $|0\rangle$), or any superposition of the two states (called a superposition state). Equation (21) describes the state of a quantum bit:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad (21)$$

where the probability amplitude of each state is determined by the complex numbers α and β , $|\alpha|^2$ and $|\beta|^2$ denote the probability of being in the “0” and “1” states within a quantum bit, which can be expressed as a two-dimensional column vector:

$$|\psi\rangle = \alpha \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (22)$$

This quantum bit follows the normalization criteria for the state representation detailed in Equations (5)–(24):

$$|\alpha|^2 + |\beta|^2 = 1 \quad (23)$$

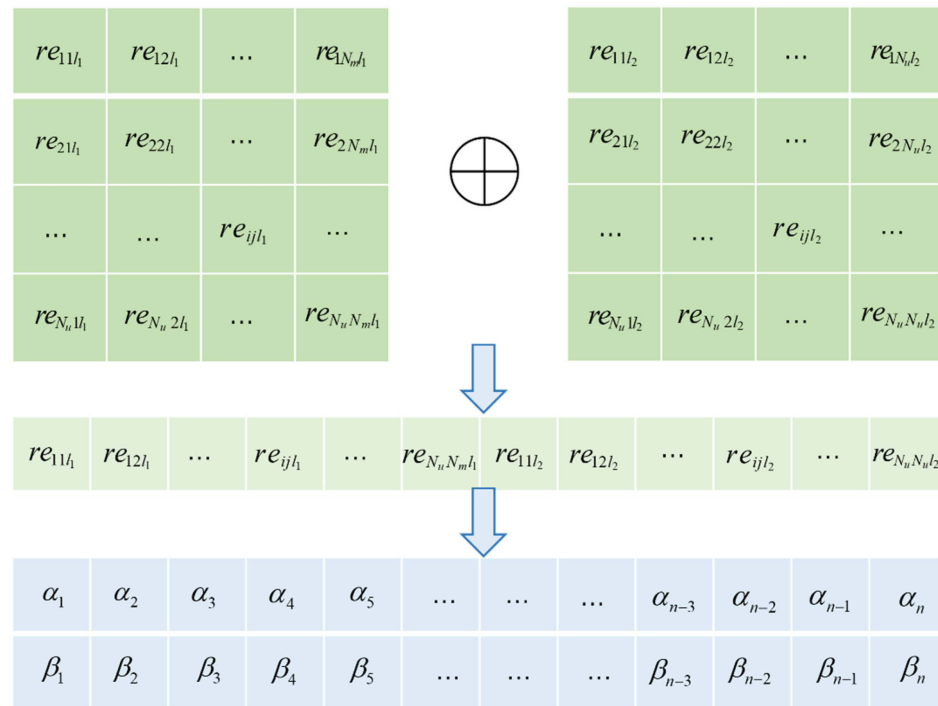


Figure 2. Quantum coding process.

Each quantum chromosome consists of a set of quantum bits, each of which represents a superposition state of a gene bit. A quantum chromosome containing n genes can be represented as

$$|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle \otimes \dots \otimes |\psi_n\rangle \quad (24)$$

At this point, each individual in the population is represented by a quantum register stored in $|\psi\rangle$ in 2^n Hilbert space, but this representation does not accurately obtain the state of a closed quantum system. Therefore, quantum measurements must be performed to evaluate the population. When measuring a quantum register, each quantum bit collapses to one of its classical states to form a classical individual, and the measurement of the i th chromosome for generation i can be expressed as Equation (25):

$$|\psi_i\rangle = \begin{bmatrix} \alpha_{i,1}^{it} \\ \beta_{i,1}^{it} \end{bmatrix} \begin{bmatrix} \alpha_{i,2}^{it} \\ \beta_{i,2}^{it} \end{bmatrix} \dots \begin{bmatrix} \alpha_{i,n}^{it} \\ \beta_{i,n}^{it} \end{bmatrix} \rightarrow [1|0|\dots|1] \quad (25)$$

3.3. Preference-Guided Quantum Gate Operating Strategy

In traditional quantum genetic algorithms, the best chromosome in the parent generation is used as an evolutionary target, and quantum gates are used to manipulate the quantum chromosomes. By rotating the quantum gate, the state of the quantum bits can be changed to update the population individuals. The probability amplitudes α and β can be equated to $\cos(\theta)$ and $\sin(\theta)$, respectively, and θ denotes the quantum bit phase. The quantum revolving door update quantum bit process is denoted as

$$\begin{bmatrix} \alpha'_i \\ \beta'_i \end{bmatrix} = \begin{bmatrix} \cos(\theta_i) & -\sin(\theta_i) \\ \sin(\theta_i) & \cos(\theta_i) \end{bmatrix} \times \begin{bmatrix} \alpha_i \\ \beta_i \end{bmatrix} \quad (26)$$

That is, the quantum bit (α_i, β_i) is converted to (α'_i, β'_i) using a quantum bit rotating gate with an angle of θ_i . The effect of the quantum gate rotation on the quantum bit operation is shown in Figure 3.

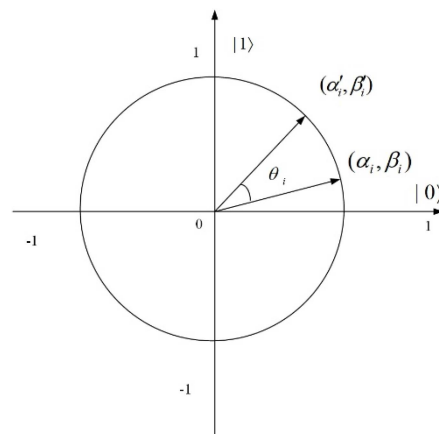


Figure 3. Schematic diagram of quantum revolving door operation.

The size of the rotation angle of the quantum gate has a very large impact on the convergence speed and accuracy: too large an angle can easily cause the algorithm to fall into the local optimal solution; too small an angle can easily lead to the convergence speed being too slow. Traditional quantum genetic algorithms all use a fixed rotation angle. An adaptive rotation angle guided by preference information has a mathematical expression shown in Equation (27):

$$\theta_i = \theta_{\max} \left(\left(\frac{f(b_{ij}) - f(x_{ij})}{f(b_{ij})} \right) + \left(\frac{f(x_{ij}) - f(p_{ij})}{f(p_{ij})} \right) \right) - (\theta_{\max} - \theta_{\min}) \frac{it}{T_{\max\text{gen}}} \quad (27)$$

where $T_{\max\text{gen}}$ is the maximum number of iterations of the algorithm, it is the number of current iteration generations, $\theta_{\max}$ is the maximum rotation angle, $\theta_{\min}$ is the minimum rotation angle, x_{ij} is the j locus of the i contemporary chromosome, b_{ij} is the j locus of the best chromosome i of the parent generation, and p_{ij} is the j locus of the i chromosome of the preferred reference point. $f(x_{ij})$, $f(b_{ij})$ and $f(p_{ij})$ are their corresponding adaptations, respectively. According to Equation (26), the size of the rotation angle is determined by the number of iterations and the difference between $f(x_{ij})$ and $f(b_{ij})$, $f(p_{ij})$.

The first half of Equation (27) selectively adjusts the rotation angle based on the characteristics of the problem and the performance of the individual to optimize results. The second half of Equation (27) shows that the rotation angle is linearly related to the number of iterations: a larger rotation angle is applied in the early stages of evolution to expand the search range, while a smaller rotation angle is used in the later stages to perform precise searches. This strategy accelerates convergence to the global optimal solution and reduces the risk of premature convergence to local optima. The rotation direction of the quantum bit is determined by the sign of $s(\alpha_i\beta_i)$. The detailed adaptive rotation angle adjustment strategy is provided in Table 2.

Table 2. Adaptive quantum gate rotation angle tuning strategy.

x_i	b_i	$f(x) \geq f(b)$	θ_i	$s(\alpha_i\beta_i)$			
				$\alpha_i\beta_i > 0$	$\alpha_i\beta_i < 0$	$\alpha_i = 0$	$\beta_i = 0$
0	0	False	θ_i	−1	+1	± 1	0
0	0	True	θ_i	−1	+1	± 1	0
0	1	False	θ_i	+1	−1	0	± 1
0	1	True	θ_i	−1	+1	± 1	0
1	0	False	θ_i	−1	+1	± 1	0
1	0	True	θ_i	+1	−1	0	± 1
1	1	False	θ_i	−1	+1	± 1	0
1	1	True	θ_i	+1	−1	0	± 1

3.4. Preferred Information Dominance Strategies

In preference-information-guided optimization, the g-dominance strategy defines a new inter-individual dominance relationship based on the decision maker's preference points, moderating the dominance relationships among individuals. The target space is divided into preference and non-preference regions, enhancing the selection pressure on the dominance ranking of individuals within the population compared to the traditional Pareto dominance algorithm, as defined below:

Definition 5. *g-dominance strategy.*

For two target vectors, $x, y \in R^m$, when one of the following conditions is satisfied, then x is said to dominate y , denoted as $x \prec_g y$:

- ① $flag_g(x) > flag_g(y)$;
- ② If $flag_g(x) = flag_g(y)$, it is necessary to satisfy any $x_i \leq y_i (\forall i = 1, 2, \dots, m)$, and there exists at least one i , such that $x_i < y_i$.

Specifically, $flag_g$ is used to divide the target space into two parts, $flag_g = 0$ and $flag_g = 1$. Individuals labeled as 1 ($flag_g = 1$) dominate individuals labeled as 0 ($flag_g = 0$), and when individuals have the same value of $flag_g$, then individuals are compared based on the Pareto dominance relationship. $flag_g$ is expressed as Equation (21), where p is a preference reference point in the objective space, which is the set of objective values consisting of the decision-maker's optimization expectation for each objective, denoted as $p(f_1^*, f_2^*, \dots, f_y^*)$.

$$flag_g(y) = \begin{cases} 1 & \text{if } y_i \leq p_i, \forall i = 1, 2, \dots, m \\ 1 & \text{if } p_i \leq y_i, \forall i = 1, 2, \dots, m \\ 0 & \text{else} \end{cases} \quad (28)$$

4. Simulation Experiment Verification and Analysis

4.1. Comparative Analysis of Algorithm Performance

4.1.1. Contrast Algorithm Selection and Parameterization

In this section, two of the most representative algorithms from the multi-objective non-dominated sorting genetic optimization category are selected: the Quantum Non-Dominated Sorting Genetic Algorithm (QNSGA-II) [7] and the g-dominance Non-Dominated Sorting Genetic Algorithm (g-NSGA-II) [8]. Comparison experiments are conducted with the PGQNSGA-II algorithm proposed in this section. The population size is set to 50, the maximum number of iterations is 200, the crossover probability is 0.9, and the mutation probability is 0.1, $\theta_{\max} = 0.05\pi$, $\theta_{\min} = 0.01\pi$. The maximum number of iterations and the maximum/minimum number of rotation angles are determined using a heuristic experimental approach. This method involves conducting a series of exploratory experiments to assess different parameter settings and selecting the ones that provided an optimal balance between computational efficiency and the algorithm's performance. The quantum gate operation of QNSGA-II is presented in Table 3, where x_{ij} and b_{ij} are the j locus of the i contemporary chromosome and the j locus of the best paternal chromosome, and $f(x)$ and $f(b)$ are their corresponding fitness, $\varepsilon = 0.02\pi$.

Table 3. Conventional quantum gate operation modes.

x_i	b_i	$f(x) \geq f(b)$	θ_i	$s(\alpha_i \beta_i)$			
				$\alpha_i \beta_i > 0$	$\alpha_i \beta_i < 0$	$\alpha_i = 0$	$\beta_i = 0$
0	0	False	0	0	0	0	0
0	0	True	0	0	0	0	0
0	1	False	ε	+1	−1	0	± 1
0	1	True	ε	−1	+1	± 1	0
1	0	False	ε	−1	+1	± 1	0
1	0	True	ε	+1	−1	0	± 1
1	1	False	0	0	0	0	0
1	1	True	0	0	0	0	0

4.1.2. Test Function Selection and Parameterization

In order to analyze the performance of the PGQNSGA-II algorithm proposed in this paper, ZDT1, ZDT2, and ZDT3 of the ZDT series and DTLZ2 of the DTLZ series of the test functions are selected. Specifically, ZDT1 is used to test basic convergence and solution uniformity, ZDT2 assesses the algorithm's handling of convex Pareto fronts, and ZDT3 introduces the challenge of discontinuous Pareto fronts. DTLZ2 is included to extend the evaluation to higher-dimensional, non-convex problems, examining the algorithm's scalability and adaptability. These functions are well-established benchmarks in the field, enabling standardized comparisons and robust performance analysis. The test function parameter settings are shown in Table 4.

Table 4. Test function parameter settings.

Function Name	Goal Quantities	Strategic Decision Quantities	Concern Causality	Preference Setting		
				Ahead	Infeasible Domain (Math.)	Feasibility Domain
ZDT1	2	30	continuous convex front	(0.42, 0.35)	(0.2, 0.2)	(0.7, 0.7)
ZDT2	2	30	continuous concave leading edge	(0.5, 0.7)	(0.2, 0.3)	(0.9, 0.8)
ZDT3	2	5	discrete discontinuous front	(0.2, 0.5)	(0.2, 0.2)	(0.6, 0.7)
DTLZ2	3	5	3D surface frontiers	(0.63, 0.38, 0.67)	(0.2, 0.4, 0.4)	(0.8, 0.8, 0.9)

4.1.3. Performance Evaluation Indicators

Generational Distance (GD) is a commonly used metric for evaluating convergence in multi-objective optimization algorithms, which indicates the degree of deviation between the solution sought by the algorithm and the true solution. The smaller the value of GD, the better the algorithm's convergence. When the value of GD is 0, the optimal solutions obtained by the table are on the real Pareto front. The calculation formula is

$$GD = \left(\frac{1}{N} \sum_{i=1}^N d_i^2 \right)^{\frac{1}{2}} \quad (29)$$

where d_i is the minimum Euclidean distance from the i th solution in the set of generated optimal solutions to the reference frontier P^* .

4.1.4. Experimental Results and Analysis

Table 5 presents the values of the Generation Distance (GD) for the algorithms being compared; specifically, it reflects the convergence of the average distance between the Pareto frontier approximation provided by each algorithm and the true Pareto optimal frontier. In this context, the QNSGA-II algorithm does not utilize preference point guidance,

rendering its GD value independent of any preference point settings. In the test results, the values highlighted in bold indicate superior performance, demonstrating that all algorithms yield highly comparable solutions. In the ZDT1 test set, the PGQNSGA-II, QNSGA-II, and g-NSGA-II algorithms demonstrate better convergence concerning the preference points across the frontier, infeasible domain, and feasible domain of ZDT1, respectively. Compared to g-NSGA-II and QNSGA-II, the PGQNSGA-II algorithm exhibits good stability in the ZDT1 test set. In two-dimensional optimization problems, the PGQNSGA-II algorithm outperforms the g-NSGA-II algorithm in preference point settings along the frontier, exhibiting greater insensitivity to these settings. The relatively weak performance of the QNSGA-II algorithm, in terms of mean and variance, can be attributed to inherent uncertainties in quantum representations; the associated noise may adversely affect the algorithm's convergence. Furthermore, the superior performance of the PGQNSGA-II algorithm, in terms of mean and variance in three-dimensional optimization problems, indicates its enhanced ability to address such problems on surfaces. In summary, the PGQNSGA-II algorithm emerges as optimal through the majority of the experimental results, exhibiting robust convergence performance, stability, and insensitivity to preference point settings across various testing scenarios involving different Pareto frontier shapes.

Table 5. Comparison of experimental results for 30 runs on the test set.

Test Function	Preference Setting	PGQNSGA-II		QNSGA-II		g-NSGA-II	
		Mean Value	Variance (Statistics)	Mean Value	Variance (Statistics)	Mean Value	Variance (Statistics)
ZDT1	ahead	<u>2.52×10^{-5}</u>	<u>2.84×10^{-5}</u>	2.22×10^{-4}	2.84×10^{-5}	4.78×10^{-5}	4.61×10^{-5}
	infeasibility horizon	2.66×10^{-5}	<u>1.66×10^{-6}</u>	2.22×10^{-4}	2.84×10^{-5}	<u>2.10×10^{-5}</u>	8.66×10^{-5}
	feasibility domain	3.52×10^{-4}	<u>8.76×10^{-6}</u>	<u>2.22×10^{-4}</u>	2.84×10^{-5}	2.25×10^{-4}	3.04×10^{-4}
ZDT2	ahead	<u>2.02×10^{-5}</u>	<u>4.88×10^{-5}</u>	2.67×10^{-3}	2.33×10^{-5}	3.22×10^{-4}	5.05×10^{-4}
	infeasibility horizon	8.31×10^{-4}	<u>7.40×10^{-5}</u>	2.67×10^{-3}	2.33×10^{-5}	<u>1.37×10^{-4}</u>	3.04×10^{-4}
	feasibility domain	6.06×10^{-4}	2.14×10^{-4}	2.67×10^{-4}	2.33×10^{-5}	<u>2.40×10^{-4}</u>	<u>1.39×10^{-4}</u>
ZDT3	ahead	<u>1.78×10^{-5}</u>	<u>4.37×10^{-5}</u>	1.22×10^{-4}	4.12×10^{-5}	2.92×10^{-5}	4.27×10^{-4}
	infeasibility horizon	<u>1.23×10^{-5}</u>	<u>8.33×10^{-5}</u>	1.22×10^{-4}	4.12×10^{-5}	2.56×10^{-5}	4.67×10^{-4}
	feasibility domain	5.38×10^{-5}	7.34×10^{-4}	1.22×10^{-4}	4.12×10^{-5}	<u>2.30×10^{-5}</u>	<u>3.89×10^{-4}</u>
DTLZ2	ahead	<u>8.66×10^{-6}</u>	<u>8.68×10^{-5}</u>	6.43×10^{-4}	6.65×10^{-5}	1.95×10^{-4}	1.60×10^{-4}
	infeasibility horizon	<u>8.72×10^{-6}</u>	<u>6.25×10^{-5}</u>	6.43×10^{-4}	6.65×10^{-5}	4.90×10^{-4}	3.60×10^{-4}
	feasibility domain	<u>2.26×10^{-6}</u>	<u>1.29×10^{-5}</u>	6.43×10^{-4}	6.65×10^{-5}	3.72×10^{-4}	3.07×10^{-4}

Values underlined in the table indicate the minimum mean and variance within each row.

4.2. Comparative Analysis of Algorithm Effectiveness

4.2.1. Scene Description and Parameterization

The MAV/UAV coordinated interception of enemy aircraft is a typical scenario in modern airspace combat. In this section, the preset combat scenario is a team of three MAVs and 10 UAVs carrying out interception and attack mission activities against eight enemy aircraft. The MAVs and UAVs carry detection loads of electro-optical infrared sensors s_1 and a number of radar sensors s_2 each, the UAVs carry one air–air proximity missile w_1 , the communication load and decision load are evaluated based on the quantity of information and the level of uncertainty it introduces, respectively, and they are quantified using an information entropy approach [21,22], as shown in Tables 6–8. The fourth column

in the Table 6 includes performance values for MAVs under different command-and-control relationships, as detailed in [20].

Table 6. Manned-machine-related parameters in the architecture.

Resource	Placement	Load Values When the Command-and-Control Relationship Is I_1	Performance Values When the Command-and-Control Relationship Is I_1
m_1	(1200, 300, 1800)	(0.95 0.87 0.67 0.45 0.29)	(0.78 0.70 0.55 0.46 0.33)
m_2	(1100, 300, 2000)	(0.90 0.85 0.59 0.40 0.30)	(0.85 0.75 0.65 0.5 0.40)
m_3	(1200, 280, 1900)	(0.93 0.86 0.60 0.44 0.30)	(0.80 0.73 0.6 0.48 0.42)

Table 7. UAV-related parameters in the architecture.

Resource	Placement	LOA Rating
u_1	(600, 300, 1800)	9
u_2	(650, 320, 1900)	7
u_3	(700, 360, 1800)	8
u_4	(600, 260, 1800)	6
u_5	(500, 200, 1800)	9
u_6	(700, 160, 1800)	5
u_7	(700, 30, 1800)	8
u_8	(800, 10, 1800)	6
u_9	(900, 30, 1800)	8
u_{10}	(800, 60, 1800)	9

Table 8. Task-related parameters in the architecture.

Task	Placement	Demand
Act_1	(100, 300, 1800)	($s_1 = 1, s_2 = 1, w_1 = 1$)
Act_2	(150, 320, 1900)	($s_1 = 1, s_2 = 1, w_1 = 1$)
Act_3	(200, 360, 1800)	($s_1 = 1, s_2 = 1, w_1 = 1$)
Act_4	(100, 260, 1800)	($s_1 = 1, s_2 = 1, w_1 = 2$)
Act_5	(100, 200, 1800)	($s_1 = 1, s_2 = 1, w_1 = 1$)
Act_6	(200, 160, 1800)	($s_1 = 1, s_2 = 1, w_1 = 2$)
Act_7	(200, 30, 1800)	($s_1 = 1, s_2 = 1, w_1 = 1$)
Act_8	(300, 10, 1800)	($s_1 = 1, s_2 = 1, w_1 = 1$)

4.2.2. Algorithm Parameterization and Results Analysis

The algorithm parameters are set as follows: initial population number of 50, maximum iteration number of 200, crossover probability uniformly set to 0.9, and mutation probability uniformly set to 0.1, $\theta_{\max} = 0.05\pi$, $\theta_{\min} = 0.01\pi$. The quantum gate operation of the QNSGA-II is the same as that shown in Table 3, in the previous section. The results of the three algorithm runs are shown in Figure 4.

Figure 4 demonstrates that the PGQNSGA-II algorithm proposed in this study achieves an architectural design scheme characterized by significantly higher total effectiveness, command-and-control performance, and execution performance when compared to the traditional QNSGA-II and g-NSGA-II algorithms. In Figure 4d, a more pronounced linear relationship is evident between system command-and-control performance and system execution performance. This observation aligns with the IPDI principle outlined in Assumption 6 and the principle that a higher degree of UAV intelligence corresponds to a reduced command-and-control load, thereby enhancing system command-and-control performance.

To further validate the performance of the PGQNSGA-II algorithm regarding convergence speed, Figure 5 illustrates the evolution of the Hyper-Volume (HV) metric concerning the number of iterations. Figure 5 indicates that the PGQNSGA-II algorithm demonstrates the fastest convergence speed, followed by the g-NSGA-II algorithm, while the

QNSGA-II algorithm exhibits inferior performance in the context of the MAV/UAV COS architecture problem.

Figure 6 presents a comparison between two MAV/UAV COS architectures, using the selected Pareto solutions (0.119, 0.967, 0.311) and (0.328, 0.728, 0.568) indicated in Figure 4. These values correspond to the objective function components: system command-and-control performance, system implementation performance, and total effectiveness. In both architectures, the MAV–UAV relationships are fully collaborative. The Pareto solution chosen for Architecture 1 is more extreme, featuring the highest system execution performance but the lowest command-and-control performance. At this point, all UAV-and-manned-aircraft relationships are directly controlled, yielding a total effectiveness of 0.31. In contrast, the Pareto solution for Architecture 2 demonstrates a more balanced approach, reflecting significant modifications. Although the execution performance of Architecture 2 decreases by 24.7% relative to Architecture 1, the command-and-control performance improves by 176.5%, and the overall effectiveness rises by 82.6%.

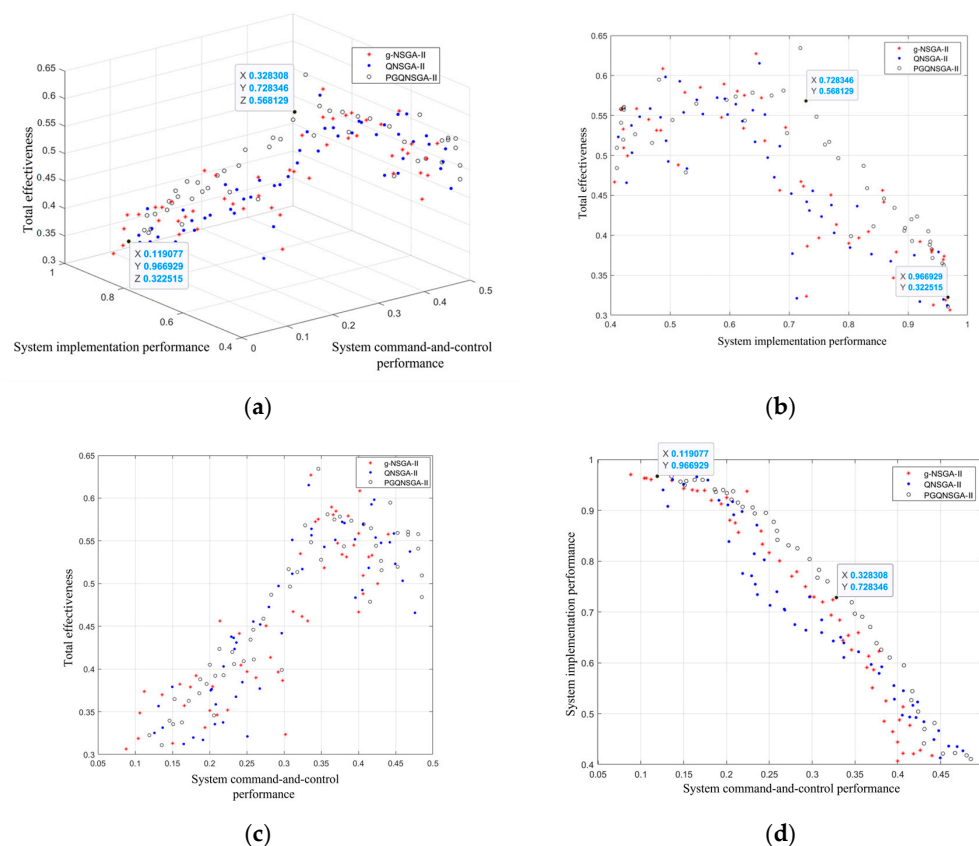


Figure 4. Comparison plot of optimized target spaces. (a) Three-dimensional target space. (b) Total effectiveness–system implementation performance target space. (c) Total effectiveness–system command and control performance target space. (d) System command-and-control performance–system execution performance target space.

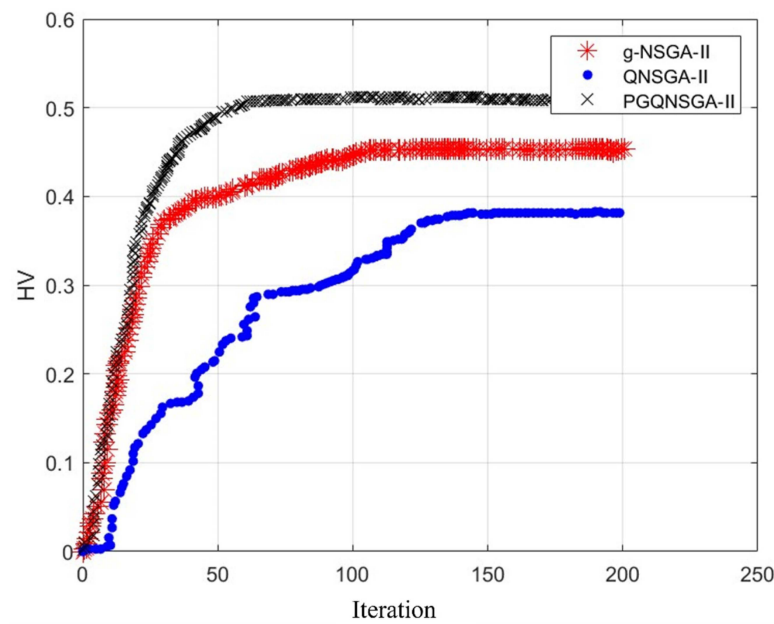


Figure 5. Curve of HV values with the number of iterations.

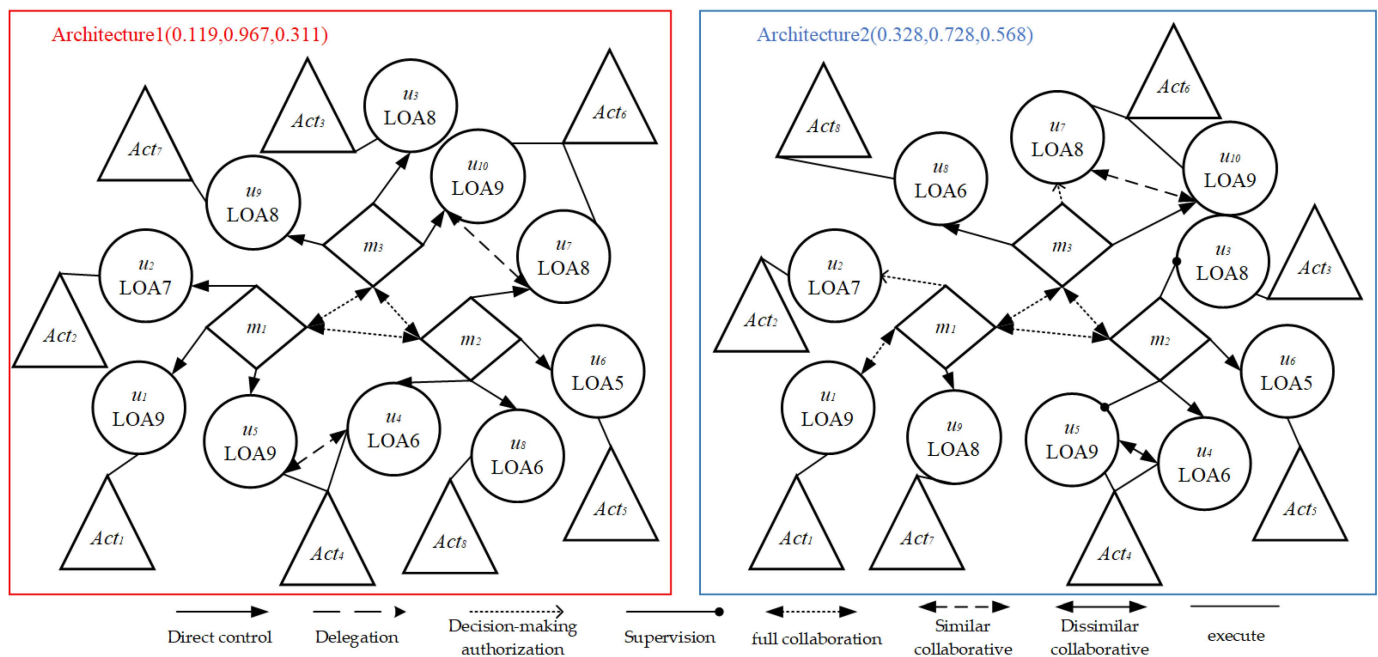


Figure 6. Comparison of MAV/UAV COS architectures.

5. Conclusions

Aiming at the optimization process of the MAV/UAV COS architecture, the decoherence of quantum bit operators in the process of multi-objective optimization will lead to premature convergence of the algorithm, thus falling into the inadequacy of the local optimum, and the PGQNSGA-II multi-objective optimization method is proposed. Based on the UAV intelligence and operator load characteristics of the MAV/UAV COS, an architectural optimization model is constructed with the optimization objectives of maximum overall system effectiveness, maximum system execution performance level, and maximum command-and-control performance, as well as the constraints of mission matching and resource limitation. A multi-objective optimization algorithm based on the combination of quantum and genetic algorithms is adopted to enhance the algorithm's

optimization search capability in the target solution space by introducing preference information to guide the quantum revolving door to adaptively select the rotation angle. By comparing the structure of simulation experiments in public test functions and real combat architecture optimization problems, the PGQNSGA-II algorithm proposed in this paper outperforms the QNSGA-II and g-NSGA-II algorithms in terms of convergence speed and performance.

In future research, we plan to extend our work to further analyze and improve the computational efficiency of the PGQNSGA-II algorithm to enhance its applicability for real-time decision-making in operational scenarios. Additionally, we will explore the feasibility of the proposed architecture optimization approach and its implementation on practical platforms to validate its effectiveness under realistic conditions. Robustness to quantum measurement errors will also be a focus, ensuring that the algorithm maintains stability and reliable performance even in the presence of noise and measurement inaccuracies. This will help solidify the method's applicability and reliability in complex, real-world manned/unmanned collaborative operations.

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References

1. Zhou, Y.; Guo, J.; Jiang, Y.; Bu, B. Design of Association Algorithm for the Manned/Unmanned airVehicle Cooperative Combat. In Proceedings of the 2017 (the Third) China Aerospace Science and Technology Conference Proceedings, Pittsburgh, PA, USA, 8–12 October 2017; pp. 392–397.
2. Klein, J.; van Vliet, H. A systematic review of system-of-systems architecture research. In *Proceedings of the 9th International ACM Sigsoft Conference on Quality of Software Architectures*, New York, NY, USA, 17–21 June 2013; Association for Computing Machinery: New York, NY, USA, 2013; pp. 13–22. [\[CrossRef\]](#)
3. Li, M.; Jia, G.; Li, X.; Qiu, H. Efficient Trajectory Planning for Optimizing Energy Consumption and Completion Time in UAV-Assisted IoT Networks. *Mathematics* **2023**, *11*, 4399. [\[CrossRef\]](#)
4. Abeywickrama, H.V.; Jayawickrama, B.A.; He, Y.; Dutkiewicz, E. Empirical Power Consumption Model for UAVs. In Proceedings of the 2018 IEEE 88th Vehicular Technology Conference (VTC-Fall), Chicago, IL, USA, 27–30 August 2018; pp. 1–5. [\[CrossRef\]](#)
5. Gunantara, N. A review of multi-objective optimization: Methods and its applications. *Cogent Eng.* **2018**, *5*, 1502242. [\[CrossRef\]](#)
6. Shu, Z.; Wang, W.; Wang, R. Design of an Optimized Architecture for Manned and Unmanned Combat System-of-Systems: Formulation and Coevolutionary Optimization. *IEEE Access* **2018**, *6*, 52725–52740. [\[CrossRef\]](#)
7. Zhou, H.; Mao, Y.; Guo, X. An Improved Multi-Objective Particle Swarm Optimization-Based Hybrid Intelligent Algorithm for Index Screening of Underwater Manned/Unmanned Cooperative System of Systems Architecture Evaluation. *Mathematics* **2023**, *11*, 4389. [\[CrossRef\]](#)
8. Han, B.; Yao, P. Coalition formation of manned/unmanned aerial vehicle cluster based on Holon organization. *Syst. Eng. Electron.* **2018**, *40*, 91–97.
9. Deutsch, J.R. Rapid solution of problems by quantum computation. *Proc. R. Soc. Lond. A* **1992**, *439*, 553–558. [\[CrossRef\]](#)
10. Olvera, C.; Montiel, O.; Rubio, Y. Quantum-inspired evolutionary algorithms on continuous space multiobjective problems. *Soft Comput.* **2023**, *27*, 13143–13164. [\[CrossRef\]](#)
11. He, L.; Wang, W. Design optimization of public building envelope based on multi-objective quantum genetic algorithm. *J. Build. Eng.* **2024**, *91*, 109714. [\[CrossRef\]](#)
12. Abraham, J.B.; Wahyunggoro, O.; Setiawan, N.A. An Effective Quantum Inspired Genetic Algorithm for Continuous Multiobjective Optimization. In Proceedings of the 2019 5th International Conference on Science in Information Technology (ICSITech), Yogyakarta, Indonesia, 23–24 October 2019; pp. 161–166. [\[CrossRef\]](#)
13. Guzel, M.; Okay, F.Y.; Kok, I.; Ozdemir, S. QNSGA-II: A Quantum Computing-Inspired Approach to Multi-Objective Optimization. In Proceedings of the 2022 International Symposium on Networks, Computers and Communications (ISNCC), Shenzhen, China, 19–22 July 2022; pp. 1–4. [\[CrossRef\]](#)
14. Khorsand, A.-R.; Wang, G.G.; Raghavan, J. Non-Dominated Sorting Genetic Quantum Algorithm for Multi-Objective Optimization. In Proceedings of the 33rd Design Automation Conference, Parts A and B, Las Vegas, NV, USA, 4–7 September 2007; Volume 6. [\[CrossRef\]](#)

15. Baklouti, A.; Dammak, K.; El Hami, A. Optimum reliable design of rolling element bearings using multi-objective optimization based on C-NSGA-II. *Reliab. Eng. Syst. Saf.* **2022**, *223*, 108508. [[CrossRef](#)]
16. Levulis, S.J.; DeLucia, P.R.; Kim, S.Y. Effects of Touch, Voice, and Multimodal Input, and Task Load on Multiple-UAV Monitoring Performance During Simulated Manned-Unmanned Teaming in a Military Helicopter. *Hum. Factors* **2018**, *60*, 1117–1129. [[CrossRef](#)] [[PubMed](#)]
17. Valavanis, K.P. The Entropy Based Approach to Modeling and Evaluating Autonomy and Intelligence of Robotic Systems. *J. Intell. Robot. Syst.* **2018**, *91*, 7–22. [[CrossRef](#)]
18. Singh, G.; Roy, R.; Chanel, C. Towards Multi-UAV and Human Interaction Driving System Exploiting Human Mental State Estimation. In Proceedings of the 12th International Joint Conference on Biomedical Engineering Systems and Technologies, Prague, Czech Republic: SCITEPRESS—Science and Technology Publications, Prague, Czech Republic, 22–24 February 2019; pp. 294–301. [[CrossRef](#)]
19. Liu, S.; Li, H. A Human Factors Integrated Methods for Weapon System Effectiveness Evaluation. In *Man-Machine-Environment System Engineering*; Long, S., Dhillon, B.S., Eds.; Springer: Singapore, 2016; pp. 469–479. [[CrossRef](#)]
20. Chen, J.; Zhang, Q.; Hou, B. An assessment method of pilot workload in manned/unmanned-aerial-vehicles team. In Proceedings of the 2017 IEEE International Conference on Signal Processing, Communications and Computing (ICSPCC), Xiamen, China, 22–25 October 2017; pp. 1–5. [[CrossRef](#)]
21. Wei, D.; Liu, X.; Li, P.; Chen, Y. Research on effectiveness evaluation method of equipment system based on node importance and improved information entropy. *Syst. Eng. Electron.* **2021**, *43*, 3614–3623. [[CrossRef](#)]
22. Luo, C.; Chen, Y.; Wang, L.; Wang, Z.; Chang, Z. Effectiveness evaluation method of system of systems based on operation loop and improved information entropy. *Syst. Eng. Electron.* **2019**, *41*, 73–80.

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